WORKED SOLUTION

STEP II 2002, Question 14

Probability and Statistics • Mixtures and Bayes' Theorem

STEP II 2002, Question 14

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/02-s2-q14

The key idea

This is a two-stage experiment: first a region is chosen, then accidents happen in it. So condition on the region. The law of total expectation gives the mean, the law of total probability gives the distribution, and Bayes' theorem turns the question round at the end. The number of accidents in a region is modelled as Poisson, because the question gives only a mean, and the target formula, with its $e^{-\lambda}\lambda^k/k!$ pattern, confirms the model.

Solution

Setting up

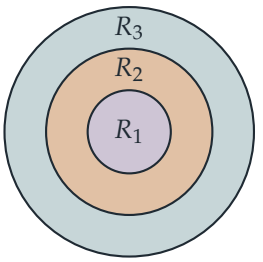
Let S be the index of the selected region and $\lambda_n = 2 - \frac{n}{N}$ the mean number of accidents in R_n. Region R_n is an annulus with radii $n-1$ and n, so its area is

$$\pi n^2 - \pi(n-1)^2 = \pi(2n-1).$$

The whole island has area πN^2. (As a check, $\sum_{n=1}^{N}(2n-1) = N^2$.) So

$$\mathrm{P}(S = n) = \frac{2n-1}{N^2}, \qquad n = 1, \ldots, N.$$

Given $S = n$, we model X as Poisson with mean λ_n. Since $\lambda_n \geq 1 > 0$ for every n, each of these Poisson distributions makes sense.



area of $R_n = \pi(2n-1)$

The long-term average

In the long run the average number recorded per day is $\mathrm{E}(X)$, by the law of large numbers. Conditioning on the region gives

$$\begin{aligned}\mathrm{E}(X) &= \sum_{n=1}^{N} \mathrm{P}(S=n)\,\mathrm{E}(X \mid S=n) = \sum_{n=1}^{N} \frac{2n-1}{N^2}\Big(2-\frac{n}{N}\Big)\\ &= 2\sum_{n=1}^{N}\frac{2n-1}{N^2} \;-\; \frac{1}{N^3}\sum_{n=1}^{N}(2n^2-n)\\ &= 2-\frac{1}{N^3}\Big(\frac{N(N+1)(2N+1)}{3}-\frac{N(N+1)}{2}\Big).\end{aligned}$$

The first sum is 1 because the selection probabilities add up to 1. In the bracket we used the given formula for $\sum n^2$ and $\sum n = \frac{1}{2}N(N+1)$. Taking out $\frac{1}{6}N(N+1)$ leaves $2(2N+1)-3 = 4N-1$, so

$$\mathrm{E}(X) = 2-\frac{(N+1)(4N-1)}{6N^2} = 2-\frac{1}{6}\Big(1+\frac{1}{N}\Big)\Big(4-\frac{1}{N}\Big),$$

splitting N^2 as $N\cdot N$ between the two factors. This is the required result.

The distribution of X

By the law of total probability, for $k = 0, 1, 2, \ldots$,

$$\mathrm{P}(X=k) = \sum_{n=1}^{N}\mathrm{P}(S=n)\,\mathrm{P}(X=k\mid S=n) = \sum_{n=1}^{N}\frac{2n-1}{N^2}\cdot\frac{\mathrm{e}^{-\lambda_n}\lambda_n^{k}}{k!}.$$

Now write each piece in terms of N:

$$\lambda_n^{k} = \Big(\frac{2N-n}{N}\Big)^{k} = N^{-k}(2N-n)^{k}, \qquad \mathrm{e}^{-\lambda_n} = \mathrm{e}^{-2+n/N} = \mathrm{e}^{-2}\,\mathrm{e}^{n/N}.$$

The factors e^{-2}, N^{-2}, N^{-k} and $1/k!$ do not depend on n, so they come out of the sum:

$$\mathrm{P}(X=k) = \frac{\mathrm{e}^{-2}N^{-k-2}}{k!}\sum_{n=1}^{N}(2n-1)(2N-n)^{k}\,\mathrm{e}^{n/N},$$

as required.

The case $N = 3$, two accidents

We want $\mathrm{P}(S=2\mid X=2)$. By Bayes' theorem,

$$\mathrm{P}(S=2\mid X=2) = \frac{\mathrm{P}(S=2)\,\mathrm{P}(X=2\mid S=2)}{\mathrm{P}(X=2)}.$$

Ask yourself Do we need to work out $\mathrm{P}(X=2)$ in full?

No. The numerator is just the $n = 2$ term of the sum for $\mathrm{P}(X = 2)$. So the common factor $\mathrm{e}^{-2}N^{-k-2}/k!$ cancels, and the probability is the $n = 2$ term of $\sum(2n-1)(2N-n)^k\mathrm{e}^{n/N}$ divided by the whole sum.

With $N = 3$ and $k = 2$, the terms $(2n-1)(6-n)^2\mathrm{e}^{n/3}$ are

$$n = 1:\ 1 \cdot 25\,\mathrm{e}^{1/3}, \qquad n = 2:\ 3 \cdot 16\,\mathrm{e}^{2/3} = 48\,\mathrm{e}^{2/3}, \qquad n = 3:\ 5 \cdot 9\,\mathrm{e} = 45\,\mathrm{e}.$$

Hence, dividing top and bottom by $\mathrm{e}^{2/3}$,

$$\mathrm{P}(S = 2 \mid X = 2) = \frac{48\,\mathrm{e}^{2/3}}{25\,\mathrm{e}^{1/3} + 48\,\mathrm{e}^{2/3} + 45\,\mathrm{e}} = \frac{48}{48 + 45\,\mathrm{e}^{1/3} + 25\,\mathrm{e}^{-1/3}},$$

which is the required result.

Checks

- For $N = 1$ there is one region with mean 1, and the formula gives $2 - \frac{1}{6} \cdot 2 \cdot 3 = 1$. As $N \to \infty$ the mean tends to $2 - \frac{4}{6} = \frac{4}{3}$. That agrees with $\int_0^1 2x(2-x)\,\mathrm{d}x = \frac{4}{3}$, where a uniform point at relative radius x has density $2x$.
- For $N = 3$ the mean is $\frac{32}{27} \approx 1.185$. A simulation of 4×10^5 days (pick a uniform point in the disc, then draw a Poisson count) gave a mean of 1.182. It also gave $\mathrm{P}(X = 2) \approx 0.209$ (exact 0.2094), and the fraction of two-accident days spent in R_2 came out at 0.372. The final answer evaluates to 0.3729.
- The distribution formula, summed over k, gives total probability 1 for every N we tried.

Common slips

- Choosing regions with equal probability $1/N$ instead of in proportion to area $2n - 1$.
- Averaging the Poisson *means* and then claiming X is Poisson with that mean. A mixture of Poissons is not Poisson. Only the expectation passes through the averaging so simply.

Beyond STEP

The variable X has a *mixture distribution*: the region acts as a hidden variable, and the last part does Bayesian inference about it. The second slip above is explained by the law of total variance, $\mathrm{Var}(X) = \mathrm{E}(\mathrm{Var}(X \mid S)) + \mathrm{Var}(\mathrm{E}(X \mid S))$, which shows that a mixture is always more spread out than a single Poisson with the same mean. Grimmett and Welsh, *Probability: An Introduction*, covers conditional expectation and these mixtures well.

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